HOME SERVICES MARKETPLACE DATABASE & APPLICATION DESIGN

Module-wise schema • columns • constraints • lookups • table mappings • application services

Stack: PostgreSQL • Java Spring Boot • React Admin • React Native Customer & Partner Apps
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1. Scope and decisions

- users contains shared account/authentication data only. first_name and last_name exist only in customer_profiles, admin_profiles, and partner_profiles.
- One user may have multiple roles via user_roles. Role-specific data belongs in profile tables.
- functions defines a module/screen/feature/API/action once; role_functions maps roles to functions with read/write/delete permissions.
- One auth_otp_requests table handles login/registration/phone-change OTPs and booking start/completion OTPs. Purpose-specific checks are mandatory.
- Customer service addresses are in addresses (multiple per customer). Admin and partner registered addresses are in their profile tables.
- Lookup tables supply centrally managed dropdown options to the UI. When a user selects an option, the business table stores the selected value_code as a raw VARCHAR string; business columns do not have foreign keys to lookup_values.
- This is a logical design specification, not a production-tested migration. Validate all migrations, provider workflows, and constraints in a development database.

2. Common audit columns

Column	Type	Rule
creation_date	TIMESTAMPTZ	NOT NULL DEFAULT now()
created_by	BIGINT	Nullable FK → users.id
updated_date	TIMESTAMPTZ	NOT NULL DEFAULT now(); application updates on every change
updated_by	BIGINT	Nullable FK → users.id
attribute1–attribute5	VARCHAR(255) each	Nullable extension fields

Apply these columns to every table below unless already listed. Prefer deactivation over hard deletion. Configure Spring Data JPA auditing; DEFAULT now() alone does not update updated_date on UPDATE.

3. Lookup tables

3.1 lookup_types

Purpose: Defines a controlled vocabulary/dropdown group.

Column	Type	Constraints / notes
id	BIGINT	PK identity
type_code	VARCHAR(80)	NN, UQ; stable code e.g. BOOKING_STATUS
type_name	VARCHAR(120)	NN
description	TEXT	Nullable
active_flag	BOOLEAN	NN DEFAULT TRUE
is_system_managed	BOOLEAN	NN DEFAULT TRUE

Constraints: Unique type_code. Include common audit columns.

Indexes: Unique type_code.

3.2 lookup_values

Purpose: Stores allowed values for each lookup group.

Column	Type	Constraints / notes
id	BIGINT	PK identity
lookup_type_id	BIGINT	NN FK lookup_types.id ON DELETE RESTRICT
value_code	VARCHAR(80)	NN
display_name	VARCHAR(150)	NN
description	TEXT	Nullable
display_order	INTEGER	NN DEFAULT 0
active_flag	BOOLEAN	NN DEFAULT TRUE

is_default	BOOLEAN	NN DEFAULT FALSE
metadata	JSONB	NN DEFAULT '{}'::jsonb

Constraints: UQ (lookup_type_id,value_code); at most one default per lookup type using partial unique index. Deactivate referenced values instead of deleting.

Indexes: (lookup_type_id,active_flag,display_order).

Lookup type code	Recommended seed values
ACCOUNT_STATUS	ACTIVE; SUSPENDED; BLOCKED; DEACTIVATED
FUNCTION_TYPE	MODULE; SCREEN; FEATURE; API; ACTION
OTP_PURPOSE	LOGIN; REGISTRATION; PHONE_CHANGE; SERVICE_START; SERVICE_COMPLETION
OTP_REQUEST_STATUS	PENDING; VERIFIED; EXPIRED; LOCKED; FAILED
ADDRESS_TYPE	HOME; WORK; REGISTERED; OTHER
PARTNER_ONBOARDING_STATUS	DRAFT; PROFILE_INCOMPLETE; UNDER_REVIEW; TRAINING_REQUIRED; APPROVED; REJECTED; SUSPENDED
VERIFICATION_STATUS	PENDING; IN_REVIEW; VERIFIED; REJECTED; EXPIRED
PARTNER_AVAILABILITY_STATUS	ONLINE; OFFLINE; BUSY
DOCUMENT_TYPE	GOVERNMENT_ID; ADDRESS_PROOF; POLICE_CHECK; CERTIFICATION; OTHER
TRAINING_STATUS	ASSIGNED; IN_PROGRESS; PASSED; FAILED
PRICING_TYPE	FIXED; HOURLY; PER_UNIT; INSPECTION
COMMISSION_TYPE	PERCENTAGE; FIXED
PRICING_RULE_TYPE	CITY_PRICE; SURGE; TIME_SLOT
PRICE_ADJUSTMENT_TYPE	FIXED; PERCENTAGE
MEDIA_TYPE	IMAGE; VIDEO
FAQ_AUDIENCE / POLICY_AUDIENCE	CUSTOMER; PARTNER; BOTH
ACCEPTANCE_METHOD	APP_CHECKBOX; WEB_CHECKBOX; SIGNED_DOCUMENT
SKILL_LEVEL	BEGINNER; INTERMEDIATE; ADVANCED; EXPERT
SKILL_ASSESSMENT_STATUS	PENDING; PASSED; FAILED; NOT_REQUIRED
BOOKING_TYPE	INSTANT; SCHEDULED
BOOKING_STATUS	PENDING; CONFIRMED; ASSIGNED; PARTNER_EN_ROUTE; IN_PROGRESS; PAUSED; COMPLETED; CANCELLED; NO_SHOW
BOOKING_PAYMENT_STATUS	PENDING; AUTHORIZED; PAID; PARTIALLY_REFUNDED; REFUNDED; FAILED; CASH_DUE
PAYMENT_METHOD	ONLINE; CASH
ASSIGNMENT_STATUS	OFFERED; ACCEPTED; REJECTED; CANCELLED; EXPIRED; COMPLETED
BOOKING_TIMER_EVENT	START; PAUSE; RESUME; STOP
SLOT_RESERVATION_STATUS	HELD; CONFIRMED; RELEASED; EXPIRED; COMPLETED
CART_STATUS	ACTIVE; CHECKED_OUT; ABANDONED
PAYMENT_ORDER_STATUS	CREATED; PENDING; AUTHORIZED; PAID; FAILED; EXPIRED; CANCELLED
PAYMENT_STATUS	PENDING; AUTHORIZED; CAPTURED; FAILED; CANCELLED; PARTIALLY_REFUNDED; REFUNDED
WEBHOOK_PROCESSING_STATUS	PENDING; PROCESSING; PROCESSED; FAILED; IGNORED
REFUND_STATUS	REQUESTED; PROCESSING; SUCCEEDED; FAILED; CANCELLED
PAYOUT_STATUS	PENDING; PROCESSING; PAID; FAILED; ON_HOLD; CANCELLED
TRANSFER_STATUS	CREATED; PROCESSING; SETTLED; FAILED; REVERSED
CASH_COLLECTION_STATUS	DUE; COLLECTED; DISPUTED; WAIVED
CANCELLATION_APPLIES_TO	CUSTOMER; PARTNER; BOTH
PENALTY_TYPE	FIXED; PERCENTAGE
CANCELLED_BY_ROLE	CUSTOMER; PARTNER; ADMIN; SYSTEM
REVIEW_STATUS	PUBLISHED; HIDDEN; UNDER_REVIEW
SUPPORT_PRIORITY	LOW; MEDIUM; HIGH; URGENT
SUPPORT_TICKET_STATUS	OPEN; ASSIGNED; IN_PROGRESS; RESOLVED; CLOSED
NOTIFICATION_CHANNEL	PUSH; SMS; EMAIL
DELIVERY_STATUS	PENDING; SENT; DELIVERED; FAILED

Business columns store the selected value_code as a raw string, with no foreign key to lookup_values. The application validates submitted codes against the relevant lookup type and active values.

4. Module 1 — Identity, authentication and access control

users

Purpose: Common account and login data; no names.

Column	Type	Constraints / notes
id	BIGINT	PK identity
country_code	VARCHAR(5)	NN
mobile_number	VARCHAR(15)	NN
email	VARCHAR(255)	Nullable; case-insensitive unique if present
profile_photo_url	TEXT	Nullable
account_status	VARCHAR(60)	NN; stores lookup value_code as raw string
account_verified_date	TIMESTAMPTZ	Nullable
status_updated_by	BIGINT	FK users.id
status_updated_date	TIMESTAMPTZ	Nullable
last_login_at	TIMESTAMPTZ	Nullable

Constraints: UQ public_id and (country_code,mobile_number). No first_name/last_name/full_name.

Indexes: (country_code,mobile_number); unique lower(email) where email is not null.

Lookup values: ACCOUNT_STATUS

roles

Purpose: Role definitions.

Column	Type	Constraints / notes
id	SMALLINT	PK identity
code	VARCHAR(30)	NN UQ
name	VARCHAR(80)	NN
description	TEXT	Nullable
active_flag	BOOLEAN	NN DEFAULT TRUE

Constraints: Seed SUPER_USER, ADMIN, SUB_ADMIN, CUSTOMER, PARTNER. roles is authoritative.

Indexes: Unique code.

user_roles

Purpose: Many-to-many role assignment.

Column	Type	Constraints / notes
user_id	BIGINT	PK part, FK users.id
role_id	SMALLINT	PK part, FK roles.id
assigned_by	BIGINT	FK users.id
assigned_at	TIMESTAMPTZ	NN DEFAULT now()
is_active	BOOLEAN	NN DEFAULT TRUE

Constraints: Composite PK (user_id,role_id); RESTRICT deletes for user and role.

Indexes: (role_id,is_active).

functions

Purpose: Defines each module/screen/feature/API/action once.

Column	Type	Constraints / notes
id	BIGINT	PK identity
function_module	VARCHAR(50)	NN e.g. PARTNER
function_code	VARCHAR(100)	NN UQ e.g. PARTNER_VERIFY
function_name	VARCHAR(150)	NN
function_type	VARCHAR(60)	NN; stores lookup value_code as raw string
description	TEXT	Nullable
active_flag	BOOLEAN	NN DEFAULT TRUE

Constraints: Do not include role_id or read/write/delete fields.

Indexes: (function_module,active_flag).

Lookup values: FUNCTION_TYPE

role_functions

Purpose: Role-specific permissions.

Column	Type	Constraints / notes
id	BIGINT	PK identity
role_id	SMALLINT	NN FK roles.id
function_id	BIGINT	NN FK functions.id
can_read	BOOLEAN	NN DEFAULT FALSE
can_write	BOOLEAN	NN DEFAULT FALSE
can_delete	BOOLEAN	NN DEFAULT FALSE
active_flag	BOOLEAN	NN DEFAULT TRUE

Constraints: UQ (role_id,function_id). Do not duplicate role/function names or codes.

Indexes: (role_id,active_flag); function_id.

auth_otp_requests

Purpose: Single table for auth and service action OTPs.

Column	Type	Constraints / notes
id	BIGINT	PK identity
user_id	BIGINT	FK users.id; nullable for pre-registration
booking_id	BIGINT	FK bookings.id; add after bookings
country_code	VARCHAR(5)	NN
mobile_number	VARCHAR(15)	NN
otp_hash	TEXT	NN; never plaintext
purpose	VARCHAR(60)	NN; stores lookup value_code as raw string
request_status	VARCHAR(60)	NN; stores lookup value_code as raw string
attempt_count	SMALLINT	NN DEFAULT 0 CHECK >=0
expires_at	TIMESTAMPTZ	NN
verified_at	TIMESTAMPTZ	Nullable

Constraints: Booking OTPs require booking_id and authorized booking/partner checks. Single-use; short expiry; rate-limit. Add partial unique pending OTP index if desired.

Indexes: (country_code,mobile_number,purpose_id,creation_date DESC); expiry index.

Lookup values: OTP_PURPOSE; OTP_REQUEST_STATUS

5. Module 2 — Profiles, addresses and partner onboarding

admin_profiles

Purpose: Admin/sub-admin profile and single registered address.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN, UQ; public-facing admin identifier, e.g. NHADMI0001
user_id	BIGINT	NN UQ FK users.id
first_name	VARCHAR(100)	NN
last_name	VARCHAR(100)	NN
employee_code	VARCHAR(30)	Nullable UQ
department	VARCHAR(100)	Nullable
job_title	VARCHAR(100)	Nullable
address_line1	VARCHAR(255)	NN
address_line2	VARCHAR(255)	Nullable

landmark	VARCHAR(255)	Nullable
city	VARCHAR(100)	NN
state	VARCHAR(100)	NN
postal_code	VARCHAR(12)	NN
country_code	VARCHAR(2)	NN DEFAULT IN

Constraints: first_name/last_name live here, not in users.

Indexes: user_id; employee_code unique when provided.

customer_profiles

Purpose: Customer personal profile and preferences.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN, UQ; public-facing customer identifier, e.g. NHCUST0001
user_id	BIGINT	NN UQ FK users.id
first_name	VARCHAR(100)	NN
last_name	VARCHAR(100)	NN
date_of_birth	DATE	Nullable
preferred_language	VARCHAR(20)	Nullable
marketing_consent	BOOLEAN	NN DEFAULT FALSE

Constraints: One profile per account; customers may own multiple addresses.

Indexes: user_id unique.

addresses

Purpose: Customer service locations.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
user_id	BIGINT	NN FK users.id
contact_name	VARCHAR(150)	NN
mobile_number	VARCHAR(15)	NN
address_type	VARCHAR(60)	NN; stores lookup value_code as raw string
label	VARCHAR(50)	Nullable
address_line1	VARCHAR(255)	NN
address_line2	VARCHAR(255)	Nullable
landmark	VARCHAR(255)	Nullable
city	VARCHAR(100)	NN
state	VARCHAR(100)	NN
postal_code	VARCHAR(12)	NN
country_code	VARCHAR(2)	NN DEFAULT IN
latitude	NUMERIC(10,7)	Nullable; -90..90
longitude	NUMERIC(10,7)	Nullable; -180..180
is_default	BOOLEAN	NN DEFAULT FALSE

Constraints: Partial UQ user_id WHERE is_default=true. Ensure address owner has CUSTOMER role in service logic.

Indexes: user_id; default-address partial unique.

Lookup values: ADDRESS_TYPE

partner_profiles

Purpose: Partner personal details, registered address and approval.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN, UQ; worker-facing ID, e.g. NHPART0001; same ID on both form pages

user_id	BIGINT	NN, UQ FK users.id; account and primary mobile number are managed in users
first_name	VARCHAR(80)	NN; Unicode given name; trim/collapse whitespace
middle_name	VARCHAR(80)	Nullable; Unicode middle name; trim/collapse whitespace
last_name	VARCHAR(80)	NN; Unicode family name; trim/collapse whitespace
aadhaar_number_encrypted	TEXT	Nullable until legal necessity, consent/legal basis and retention are approved; encrypt, mask, tightly restrict and never log
alternate_contact_number	VARCHAR(10)	Nullable; Indian mobile number; must differ from primary mobile if supplied
is_adult	BOOLEAN	NN; must be TRUE before approval/activation; self-declaration is not age proof
preferred_contact_language	VARCHAR(40)	NN; stores configured language value_code as raw string, e.g. te, hi, en
address_line1	VARCHAR(300)	NN; current address
address_line2	VARCHAR(255)	Nullable
landmark	VARCHAR(255)	Nullable
locality	VARCHAR(100)	NN; area/locality text or configured locality code
city	VARCHAR(100)	NN
state	VARCHAR(100)	NN
postal_code	CHAR(6)	NN; Indian PIN code; preserve as string
country_code	VARCHAR(2)	NN DEFAULT IN
latitude	NUMERIC(10,7)	Nullable; -90 to 90
longitude	NUMERIC(10,7)	Nullable; -180 to 180
registration_date	DATE	NN DEFAULT CURRENT_DATE; editable only by authorised staff
registered_by	BIGINT	NN FK users.id; authenticated staff account, not free text
skill_category	VARCHAR(30)	NN; value_code skilled or unskilled
years_experience	SMALLINT	NN; proposed range 0–60
main_service_category_id	BIGINT	NN FK service_categories.id
additional_skills	JSONB	Nullable array of configured skill codes/IDs; deduplicate and validate each item
preferred_arrangement	VARCHAR(40)	NN; value_code FLEXIBLE_JOBS, COMMITTED_PART_TIME, CORE_FULL_TIME
payment_basis	VARCHAR(20)	NN; value_code PER_JOB, PER_HOUR, PER_DAY, MONTHLY
expected_earnings_inr	NUMERIC(10,2)	NN; > 0; unit is payment_basis
visit_charge_inr	NUMERIC(10,2)	Nullable; >= 0
minimum_job_payment_inr	NUMERIC(10,2)	Nullable; >= 0
previous_work_details	TEXT	Nullable; previous work experience summary
service_areas	JSONB	NN; array of configured area IDs/codes or names; no partner_service_areas table
max_travel_km	NUMERIC(5,1)	NN; proposed range 0–999.9; consider city-level cap
weekly_availability	JSONB	NN; selected day/slot combinations, 7 days × 5 slots; validate codes; at least

		one slot recommended
weekend_availability	VARCHAR(20)	NN; value_code SATURDAY, SUNDAY, NEITHER, VARIES; reconcile with weekly grid
emergency_contact_name	VARCHAR(120)	Nullable until required-field policy is confirmed
emergency_contact_relationship	VARCHAR(60)	Nullable; configured relationship code or OTHER
emergency_contact_number	VARCHAR(10)	Nullable; Indian mobile number
emergency_contact_permission	BOOLEAN	NN DEFAULT FALSE; do not share/activate emergency contact unless worker confirms permission
onboarding_status	VARCHAR(60)	NN; stores lookup value_code as raw string
verification_status	VARCHAR(60)	NN; stores lookup value_code as raw string
availability_status	VARCHAR(60)	NN; stores lookup value_code as raw string
approved_by	BIGINT	Nullable FK users.id
approved_at	TIMESTAMPTZ	Nullable
rejection_reason	TEXT	Nullable
creation_date	TIMESTAMPTZ	NN DEFAULT now()
created_by	BIGINT	Nullable FK users.id
updated_date	TIMESTAMPTZ	NN DEFAULT now()
updated_by	BIGINT	Nullable FK users.id
attribute1–attribute5	VARCHAR(255)	Optional extension fields, consistent with common audit convention

Constraints: If approved_at is set, approved_by is required. Require is_adult=TRUE before approval; do not share emergency contact unless permission is TRUE. Aadhaar collection remains subject to approved legal basis, consent and retention policy. Validate all profile fields on client and server; allow draft saves and enforce required fields at final submission.

Indexes: (verification_status,onboarding_status); (availability_status,city); unique public_id; user_id unique.

Lookup values: PARTNER_ONBOARDING_STATUS; VERIFICATION_STATUS; PARTNER_AVAILABILITY_STATUS; PREFERRED_CONTACT_LANGUAGE; SKILL_CATEGORY; PREFERRED_ARRANGEMENT; PAYMENT_BASIS; WEEKEND_AVAILABILITY; EMERGENCY_CONTACT_RELATIONSHIP

partner_services

Purpose: Maps each partner to the services they are qualified and willing to perform, with per-service eligibility and optional partner-specific experience or pricing notes.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
service_id	BIGINT	NN FK services.id
experience_years	SMALLINT	Nullable; 0–60
is_primary_service	BOOLEAN	NN DEFAULT FALSE
is_active	BOOLEAN	NN DEFAULT TRUE
created_at	TIMESTAMPTZ	NN DEFAULT now()
updated_at	TIMESTAMPTZ	NN DEFAULT now()

partner_documents

Purpose: Partner identity/certification records.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
document_type	VARCHAR(60)	NN; stores lookup value_code as raw

		string
document_number_encrypted	TEXT	Nullable encrypted
file_storage_key	TEXT	NN
verification_status	VARCHAR(60)	NN; stores lookup value_code as raw string
verified_by	BIGINT	FK users.id
verified_at	TIMESTAMPTZ	Nullable
expiry_date	DATE	Nullable
rejection_reason	TEXT	Nullable

Constraints: Use protected object storage; do not expose document storage keys publicly.

Indexes: (partner_id,verification_status_id).

Lookup values: DOCUMENT_TYPE; VERIFICATION_STATUS

partner_bank_accounts

Purpose: Partner payout account information.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
account_holder_name	VARCHAR(150)	NN
account_number_encrypted	TEXT	NN encrypted
account_number_last4	VARCHAR(4)	NN exactly four digits
ifsc_code	VARCHAR(11)	NN IFSC pattern
bank_name	VARCHAR(150)	Nullable
verification_status	VARCHAR(60)	NN; stores lookup value_code as raw string
gateway_beneficiary_id	VARCHAR(150)	Nullable
is_primary	BOOLEAN	NN DEFAULT TRUE

Constraints: Partial UQ partner_id WHERE is_primary=true; restrict access to payout service.

Indexes: (partner_id); primary partial unique.

Lookup values: VERIFICATION_STATUS

6. Module 3 — Service catalogue, pricing and policies

service_categories

Purpose: Top-level service categories.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
name	VARCHAR(120)	NN
slug	VARCHAR(150)	NN UQ
description	TEXT	Nullable
icon_url	TEXT	Nullable
display_order	INTEGER	NN DEFAULT 0
is_active	BOOLEAN	NN DEFAULT TRUE

Constraints: Deactivate categories already used by services.

Indexes: (display_order,name) WHERE is_active.

services

Purpose: Service catalogue and default price.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ

category_id	BIGINT	NN FK service_categories.id
name	VARCHAR(150)	NN
slug	VARCHAR(180)	NN UQ
short_description	VARCHAR(300)	Nullable
description	TEXT	Nullable
pricing_type	VARCHAR(60)	NN; stores lookup value_code as raw string
base_price	NUMERIC(12,2)	NN >=0
minimum_charge	NUMERIC(12,2)	NN DEFAULT 0 >=0
estimated_duration_minutes	INTEGER	>0 if present
minimum_billable_minutes	INTEGER	>0 if present
billing_increment_minutes	INTEGER	>0 if present
platform_commission_type	VARCHAR(60)	NN; stores lookup value_code as raw string
platform_commission_value	NUMERIC(8,2)	NN DEFAULT 0 >=0; percentage <=100
is_instant_booking_allowed	BOOLEAN	NN DEFAULT TRUE
is_scheduled_booking_allowed	BOOLEAN	NN DEFAULT TRUE
is_active	BOOLEAN	NN DEFAULT TRUE
Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
service_id	BIGINT	NN FK services.id
variant_code	VARCHAR(80)	NN; unique within service
name	VARCHAR(150)	NN
description	TEXT	Nullable
price_override	NUMERIC(12,2)	Nullable >=0; if null, use service pricing
duration_minutes	INTEGER	Nullable >0; overrides service estimate when supplied
display_order	INTEGER	NN DEFAULT 0
is_active	BOOLEAN	NN DEFAULT TRUE

Constraints: Hourly pricing requires valid minimum and billing increment. Snapshot price at booking creation.

Indexes: (category_id,name); pricing type where active.

Lookup values: PRICING_TYPE; COMMISSION_TYPE

service_media

Purpose: Service images/videos.

Column	Type	Constraints / notes
id	BIGINT	PK identity
service_id	BIGINT	NN FK services.id
media_type	VARCHAR(60)	NN; stores lookup value_code as raw string
storage_key	TEXT	NN
display_order	INTEGER	NN DEFAULT 0
is_active	BOOLEAN	NN DEFAULT TRUE

Constraints: Use CDN/private storage according to asset type.

Indexes: (service_id,display_order) WHERE active.

Lookup values: MEDIA_TYPE

service_variants

Purpose: Service package variants.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN, UQ; public-facing variant identifier
service_id	BIGINT	NN FK services.id
variant_code	VARCHAR(80)	NN; unique within service

name	VARCHAR(150)	NN
description	TEXT	Nullable
price_override	NUMERIC(12,2)	Nullable; must be >= 0; if NULL, use service default pricing
duration_minutes	INTEGER	Nullable; must be > 0; overrides service estimate when supplied
display_order	INTEGER	NN DEFAULT 0
is_active	BOOLEAN	NN DEFAULT TRUE

Constraints: Validate that selected variant belongs to the selected service.

Indexes: (service_id,is_active).

service_requirements

Purpose: Customer prerequisites/instructions.

Column	Type	Constraints / notes
id	BIGINT	PK identity
service_id	BIGINT	NN FK services.id
requirement_type	VARCHAR(30)	NN; use lookup if configurable
description	TEXT	NN
is_mandatory	BOOLEAN	NN DEFAULT TRUE
display_order	INTEGER	NN DEFAULT 0

Constraints: Use lookup for requirement_type if admins manage the options.

Indexes: (service_id,display_order).

faqs

Purpose: FAQs for customer/partner/category/service.

Column	Type	Constraints / notes
id	BIGINT	PK identity
question	VARCHAR(500)	NN
answer	TEXT	NN
category_id	BIGINT	FK service_categories.id
service_id	BIGINT	FK services.id
audience	VARCHAR(60)	NN; stores lookup value_code as raw string
display_order	INTEGER	NN DEFAULT 0
is_published	BOOLEAN	NN DEFAULT FALSE

Constraints: At most one of category_id/service_id populated.

Indexes: (audience_id,display_order) WHERE published.

Lookup values: FAQ_AUDIENCE

policy_documents

Purpose: Versioned terms/policies.

Column	Type	Constraints / notes
id	BIGINT	PK identity
policy_key	VARCHAR(80)	NN
title	VARCHAR(200)	NN
audience	VARCHAR(60)	NN; stores lookup value_code as raw string
version	VARCHAR(30)	NN
content	TEXT	NN
effective_from	TIMESTAMPTZ	NN
published_at	TIMESTAMPTZ	Nullable
is_published	BOOLEAN	NN DEFAULT FALSE

Constraints: UQ (policy_key,version); preserve historical versions.

Indexes: (policy_key,effective_from DESC) WHERE published.

Lookup values: POLICY_AUDIENCE

policy_acceptances

Purpose: User acceptance records.

Column	Type	Constraints / notes
id	BIGINT	PK identity
user_id	BIGINT	NN FK users.id
policy_id	BIGINT	NN FK policy_documents.id
accepted_at	TIMESTAMPTZ	NN DEFAULT now()
acceptance_method	VARCHAR(60)	NN; stores lookup value_code as raw string
ip_address	INET	Nullable

Constraints: UQ (user_id,policy_id) if only latest acceptance per policy is stored.

Indexes: (user_id,accepted_at DESC).

Lookup values: ACCEPTANCE_METHOD

7. Module 4 — Partner skills

partner_skills

Purpose: Service eligibility and assessment per partner.

Column	Type	Constraints / notes
partner_id	BIGINT	PK part, FK partner_profiles.id
service_id	BIGINT	PK part, FK services.id
skill_level	VARCHAR(60)	stores lookup value_code as raw string
assessment_status	VARCHAR(60)	NN; stores lookup value_code as raw string
assessed_by	BIGINT	FK users.id
assessed_at	TIMESTAMPTZ	Nullable

Constraints: Composite PK (partner_id,service_id). Only eligible partners may be assigned.

Indexes: (service_id,assessment_status_id).

Lookup values: SKILL_LEVEL; SKILL_ASSESSMENT_STATUS

8. Module 5 — Availability and slots

partner_availability

Purpose: Weekly recurring working hours.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
day_of_week	SMALLINT	NN 1–7 ISO weekday
start_time	TIME	NN
end_time	TIME	NN
timezone	VARCHAR(60)	NN DEFAULT Asia/Kolkata
is_available	BOOLEAN	NN DEFAULT TRUE

Constraints: start_time != end_time; define overnight shifts explicitly.

Indexes: (partner_id,day_of_week,is_available).

partner_time_off

Purpose: Leave and unavailable periods.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
starts_at	TIMESTAMPTZ	NN
ends_at	TIMESTAMPTZ	NN
reason	VARCHAR(200)	Nullable

Constraints: ends_at > starts_at.

Indexes: (partner_id,starts_at,ends_at).

partner_slot_reservations

Purpose: Held/confirmed booking slots.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
booking_id	BIGINT	FK bookings.id; add after bookings
starts_at	TIMESTAMPTZ	NN
ends_at	TIMESTAMPTZ	NN
reservation_status	VARCHAR(60)	NN; stores lookup value_code as raw string
hold_expires_at	TIMESTAMPTZ	Nullable

Constraints: ends_at > starts_at. Use btree_gist exclusion constraint to prevent overlapping HELD/CONFIRMED reservations for one partner.

Indexes: (reservation_status_id,starts_at).

Lookup values: SLOT_RESERVATION_STATUS

9. Module 6 — Cart, bookings and tracking

carts

Purpose: Customer shopping cart.

Column	Type	Constraints / notes
id	BIGINT	PK identity
customer_id	BIGINT	NN FK customer_profiles.id
status	VARCHAR(60)	NN; stores lookup value_code as raw string

Constraints: One active cart per customer using partial unique index.

Indexes: customer_id where ACTIVE.

Lookup values: CART_STATUS

cart_items

Purpose: Service and options in cart.

Column	Type	Constraints / notes
id	BIGINT	PK identity
cart_id	BIGINT	NN FK carts.id
service_id	BIGINT	NN FK services.id
variant_id	BIGINT	FK service_variants.id
estimated_unit_price	NUMERIC(12,2)	NN >=0; informational only

Constraints: Recalculate prices server-side at checkout.

Indexes: (cart_id).

bookings

Purpose: Main service transaction and immutable snapshots.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
checkout_id	UUID	Nullable; groups checkout bookings
customer_id	BIGINT	NN FK customer_profiles.id
service_id	BIGINT	NN FK services.id
variant_id	BIGINT	FK service_variants.id
partner_id	BIGINT	FK partner_profiles.id
address_id	BIGINT	NN FK addresses.id
booking_type	VARCHAR(60)	NN; stores lookup value_code as raw string
booking_status	VARCHAR(60)	NN; stores lookup value_code as raw string
payment_status	VARCHAR(60)	NN; stores lookup value_code as raw string
payment_method	VARCHAR(60)	NN; stores lookup value_code as raw string
scheduled_start_at	TIMESTAMPTZ	NN
scheduled_end_at	TIMESTAMPTZ	Nullable > start
actual_start_at	TIMESTAMPTZ	Nullable
actual_end_at	TIMESTAMPTZ	Nullable > actual start
service_name_snapshot	VARCHAR(150)	NN
address_snapshot	JSONB	NN
pricing_snapshot	JSONB	NN
subtotal_amount	NUMERIC(12,2)	NN >=0
discount_amount	NUMERIC(12,2)	NN DEFAULT 0 >=0
tax_amount	NUMERIC(12,2)	NN DEFAULT 0 >=0
total_amount	NUMERIC(12,2)	NN >=0
currency	CHAR(3)	NN DEFAULT INR
cancellation_reason	TEXT	Nullable

Constraints: Validate state transitions in backend; use transactions. Index customer+date, partner+status+schedule, status+date, checkout_id, scheduled_start_at. Snapshots preserve price/address.

Indexes: (customer_id,creation_date DESC); (partner_id,booking_status_id,scheduled_start_at); (booking_status_id,creation_date); checkout_id.

Lookup values: BOOKING_TYPE; BOOKING_STATUS; BOOKING_PAYMENT_STATUS; PAYMENT_METHOD

booking_items

Purpose: Line items and price snapshots.

Column	Type	Constraints / notes
id	BIGINT	PK identity
booking_id	BIGINT	NN FK bookings.id
service_id	BIGINT	NN FK services.id
variant_id	BIGINT	FK service_variants.id
item_name_snapshot	VARCHAR(150)	NN
unit_price	NUMERIC(12,2)	NN >=0
line_total	NUMERIC(12,2)	NN >=0

Constraints: Index booking_id; keep snapshots after catalogue changes.

Indexes: booking_id.

booking_status_history

Purpose: Audit of booking state changes.

Column	Type	Constraints / notes
id	BIGINT	PK identity

booking_id	BIGINT	NN FK bookings.id
old_status	VARCHAR(60)	stores lookup value_code as raw string
new_status	VARCHAR(60)	NN; stores lookup value_code as raw string
changed_by	BIGINT	FK users.id
reason	TEXT	Nullable

Constraints: Insert in same transaction as booking update.

Indexes: (booking_id,creation_date).

Lookup values: BOOKING_STATUS

booking_assignments

Purpose: Manual assignment attempts and acceptance.

Column	Type	Constraints / notes
id	BIGINT	PK identity
booking_id	BIGINT	NN FK bookings.id
partner_id	BIGINT	NN FK partner_profiles.id
assigned_by	BIGINT	FK users.id
assignment_status	VARCHAR(60)	NN; stores lookup value_code as raw string
offered_at	TIMESTAMPTZ	NN DEFAULT now()
responded_at	TIMESTAMPTZ	Nullable
expires_at	TIMESTAMPTZ	Nullable
rejection_reason	TEXT	Nullable

Constraints: Partial unique index one ACCEPTED assignment per booking; validate partner eligibility.

Indexes: (partner_id,assignment_status_id); accepted assignment partial unique.

Lookup values: ASSIGNMENT_STATUS

booking_time_logs

Purpose: Service timer event history.

Column	Type	Constraints / notes
id	BIGINT	PK identity
booking_id	BIGINT	NN FK bookings.id
event_type	VARCHAR(60)	NN; stores lookup value_code as raw string
event_at	TIMESTAMPTZ	NN DEFAULT now()
recorded_by	BIGINT	FK users.id
notes	TEXT	Nullable

Constraints: Validate event sequence; calculate billable time on backend.

Indexes: (booking_id,event_at).

Lookup values: BOOKING_TIMER_EVENT

partner_location_updates

Purpose: Partner location history for live tracking.

Column	Type	Constraints / notes
id	BIGINT	PK identity
partner_id	BIGINT	NN FK partner_profiles.id
booking_id	BIGINT	FK bookings.id
latitude	NUMERIC(10,7)	NN; -90..90
longitude	NUMERIC(10,7)	NN; -180..180
recorded_at	TIMESTAMPTZ	NN DEFAULT now()

Constraints: Restrict visibility to booking participants; apply retention/partitioning if volume grows.

Indexes: (partner_id,booking_id,recorded_at DESC).

10. Module 7 — Payments, commissions and payouts

payment_orders

Purpose: Gateway order for a checkout.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
checkout_id	UUID	NN
customer_id	BIGINT	NN FK customer_profiles.id
provider	VARCHAR(30)	NN
provider_order_id	VARCHAR(150)	Nullable
amount	NUMERIC(12,2)	NN >0
currency	CHAR(3)	NN DEFAULT INR
payment_status	VARCHAR(60)	NN; stores lookup value_code as raw string
idempotency_key	VARCHAR(100)	NN UQ
expires_at	TIMESTAMPTZ	Nullable

Constraints: UQ (provider,provider_order_id) when supplied. Do not store card credentials.

Indexes: checkout_id; customer/date.

Lookup values: PAYMENT_ORDER_STATUS

payments

Purpose: Payment attempts/captures per booking.

Column	Type	Constraints / notes
id	BIGINT	PK identity
payment_order_id	BIGINT	NN FK payment_orders.id
booking_id	BIGINT	NN FK bookings.id
provider	VARCHAR(30)	NN
provider_payment_id	VARCHAR(150)	Nullable
payment_method	VARCHAR(60)	NN; stores lookup value_code as raw string
amount	NUMERIC(12,2)	NN >0
currency	CHAR(3)	NN DEFAULT INR
status	VARCHAR(60)	NN; stores lookup value_code as raw string
failure_code	VARCHAR(100)	Nullable
failure_message	TEXT	Nullable
captured_at	TIMESTAMPTZ	Nullable

Constraints: Unique provider payment ID when present; index by booking and status.

Indexes: payment_order_id; booking_id; status/date.

Lookup values: PAYMENT_METHOD; PAYMENT_STATUS

payment_webhook_events

Purpose: Idempotent provider webhook inbox.

Column	Type	Constraints / notes
id	BIGINT	PK identity
provider	VARCHAR(30)	NN
provider_event_id	VARCHAR(150)	NN
event_type	VARCHAR(100)	NN
payload_json	JSONB	NN
signature_verified	BOOLEAN	NN DEFAULT FALSE
processing_status	VARCHAR(60)	NN; stores lookup value_code as raw string

received_at	TIMESTAMPTZ	NN DEFAULT now()
processed_at	TIMESTAMPTZ	Nullable

Constraints: UQ (provider,provider_event_id); verify signature before processing.

Indexes: (processing_status_id,received_at).

Lookup values: WEBHOOK_PROCESSING_STATUS

payment_refunds

Purpose: Refund requests and outcomes.

Column	Type	Constraints / notes
id	BIGINT	PK identity
payment_id	BIGINT	NN FK payments.id
booking_id	BIGINT	NN FK bookings.id
provider_refund_id	VARCHAR(150)	Nullable
amount	NUMERIC(12,2)	NN >0
reason	TEXT	Nullable
status	VARCHAR(60)	NN; stores lookup value_code as raw string
requested_by	BIGINT	FK users.id
completed_at	TIMESTAMPTZ	Nullable

Constraints: Successful refund total must not exceed captured amount; enforce transactionally.

Indexes: (booking_id,status_id).

Lookup values: REFUND_STATUS

commission_rules

Purpose: Commission percentage/fixed amount rules.

Column	Type	Constraints / notes
id	BIGINT	PK identity
service_id	BIGINT	FK services.id
category_id	BIGINT	FK service_categories.id
commission_type	VARCHAR(60)	NN; stores lookup value_code as raw string
commission_value	NUMERIC(8,2)	NN >=0; percentage <=100
valid_from	TIMESTAMPTZ	NN DEFAULT now()
valid_until	TIMESTAMPTZ	Nullable
is_active	BOOLEAN	NN DEFAULT TRUE

Constraints: Define service-vs-category precedence and valid date checks.

Indexes: (is_active,service_id,category_id,valid_from).

Lookup values: COMMISSION_TYPE

partner_payouts

Purpose: Partner earning and payable ledger per booking.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
partner_id	BIGINT	NN FK partner_profiles.id
booking_id	BIGINT	NN FK bookings.id
payment_id	BIGINT	FK payments.id
gross_amount	NUMERIC(12,2)	NN >=0
platform_commission	NUMERIC(12,2)	NN >=0
adjustments_amount	NUMERIC(12,2)	NN DEFAULT 0
net_amount	NUMERIC(12,2)	NN >=0
provider_transfer_id	VARCHAR(150)	Nullable
payout_status	VARCHAR(60)	NN; stores lookup value_code as raw string

scheduled_at	TIMESTAMPTZ	Nullable
paid_at	TIMESTAMPTZ	Nullable

Constraints: Define if multiple payout adjustments per booking are allowed; reconcile net amount.

Indexes: (partner_id,payout_status_id,creation_date DESC).

Lookup values: PAYOUT_STATUS

partner_transfers

Purpose: Transfer attempts to partner bank/gateway account.

Column	Type	Constraints / notes
id	BIGINT	PK identity
payout_id	BIGINT	NN FK partner_payouts.id
provider_transfer_id	VARCHAR(150)	Nullable UQ
amount	NUMERIC(12,2)	NN >0
status	VARCHAR(60)	NN; stores lookup value_code as raw string
idempotency_key	VARCHAR(100)	NN UQ
provider_response	JSONB	Nullable
settled_at	TIMESTAMPTZ	Nullable

Constraints: Do not log provider secrets; retry with same idempotency key.

Indexes: (status_id,creation_date).

Lookup values: TRANSFER_STATUS

cash_collections

Purpose: Cash collected by partner for booking.

Column	Type	Constraints / notes
id	BIGINT	PK identity
booking_id	BIGINT	NN UQ FK bookings.id
amount	NUMERIC(12,2)	NN >=0
status	VARCHAR(60)	NN; stores lookup value_code as raw string
collected_by	BIGINT	FK users.id
confirmed_at	TIMESTAMPTZ	Nullable
customer_acknowledged_at	TIMESTAMPTZ	Nullable
notes	TEXT	Nullable

Constraints: If COLLECTED, collected_by and confirmed_at required.

Indexes: (status_id,creation_date).

Lookup values: CASH_COLLECTION_STATUS

11. Module 8 — Cancellation

cancellation_policies

Purpose: Cancellation rules, optionally service-specific.

Column	Type	Constraints / notes
id	BIGINT	PK identity
name	VARCHAR(150)	NN
applies_to	VARCHAR(60)	NN; stores lookup value_code as raw string
service_id	BIGINT	FK services.id; null for general policy
free_cancellation_minutes	INTEGER	NN DEFAULT 0 >=0
penalty_type	VARCHAR(60)	NN; stores lookup value_code as raw string

penalty_value	NUMERIC(12,2)	NN >=0; percentage <=100
is_active	BOOLEAN	NN DEFAULT TRUE
valid_from	TIMESTAMPTZ	NN DEFAULT now()
valid_until	TIMESTAMPTZ	Nullable

Constraints: valid_until > valid_from when present; define policy precedence.

Indexes: (is_active,service_id,valid_from).

Lookup values: CANCELLATION_APPLIES_TO; PENALTY_TYPE

booking_cancellations

Purpose: Applied cancellation details and policy snapshot.

Column	Type	Constraints / notes
id	BIGINT	PK identity
booking_id	BIGINT	NN UQ FK bookings.id
cancelled_by	BIGINT	FK users.id
cancelled_by_role	VARCHAR(60)	NN; stores lookup value_code as raw string
cancellation_policy_id	BIGINT	FK cancellation_policies.id
policy_snapshot	JSONB	NN DEFAULT {}
reason_code	VARCHAR(60)	Nullable
reason_text	TEXT	Nullable
penalty_amount	NUMERIC(12,2)	NN DEFAULT 0 >=0
refund_amount	NUMERIC(12,2)	NN DEFAULT 0 >=0

Constraints: Policy snapshot records the exact rules applied.

Indexes: booking_id unique.

Lookup values: CANCELLED_BY_ROLE

12. Module 9 — Reviews, support, notifications and audit

booking_reviews

Purpose: Customer rating for a completed booking.

Column	Type	Constraints / notes
id	BIGINT	PK identity
booking_id	BIGINT	NN UQ FK bookings.id
customer_id	BIGINT	NN FK customer_profiles.id
partner_id	BIGINT	NN FK partner_profiles.id
service_rating	SMALLINT	NN 1..5
partner_rating	SMALLINT	NN 1..5
review_text	TEXT	Nullable
review_status	VARCHAR(60)	NN; stores lookup value_code as raw string

Constraints: Only eligible completed bookings may be reviewed.

Indexes: (partner_id,partner_rating); service_rating.

Lookup values: REVIEW_STATUS

support_tickets

Purpose: Support requests.

Column	Type	Constraints / notes
id	BIGINT	PK identity
public_id	VARCHAR(24)	NN UQ
requester_id	BIGINT	NN FK users.id
booking_id	BIGINT	FK bookings.id
payment_id	BIGINT	FK payments.id

category	VARCHAR(40)	NN; use lookup if configurable
subject	VARCHAR(200)	NN
description	TEXT	NN
priority	VARCHAR(60)	NN; stores lookup value_code as raw string
status	VARCHAR(60)	NN; stores lookup value_code as raw string
assigned_to	BIGINT	FK users.id
resolved_at	TIMESTAMPTZ	Nullable
closed_at	TIMESTAMPTZ	Nullable

Constraints: requester_id is ticket owner; created_by is audit actor.

Indexes: (status_id,priority_id,creation_date); booking_id.

Lookup values: SUPPORT_PRIORITY; SUPPORT_TICKET_STATUS

support_ticket_messages

Purpose: Ticket messages and internal notes.

Column	Type	Constraints / notes
id	BIGINT	PK identity
ticket_id	BIGINT	NN FK support_tickets.id
sender_id	BIGINT	NN FK users.id
message	TEXT	NN
is_internal_note	BOOLEAN	NN DEFAULT FALSE

Constraints: Only authorized staff can view internal notes.

Indexes: (ticket_id,creation_date).

support_ticket_attachments

Purpose: Files attached to ticket or message.

Column	Type	Constraints / notes
id	BIGINT	PK identity
ticket_id	BIGINT	NN FK support_tickets.id
message_id	BIGINT	FK support_ticket_messages.id
storage_key	TEXT	NN
file_type	VARCHAR(100)	NN

Constraints: Private object storage; validate size, MIME type and access.

Indexes: (ticket_id).

notifications

Purpose: In-app notification.

Column	Type	Constraints / notes
id	BIGINT	PK identity
user_id	BIGINT	NN FK users.id
notification_type	VARCHAR(50)	NN; lookup if configurable
title	VARCHAR(200)	NN
message	TEXT	NN
reference_type	VARCHAR(40)	Nullable
reference_id	BIGINT	Nullable polymorphic ID
read_at	TIMESTAMPTZ	Nullable

Constraints: Polymorphic reference is validated in service logic.

Indexes: (user_id,creation_date DESC).

notification_delivery_logs

Purpose: Per-channel delivery status.

Column	Type	Constraints / notes
id	BIGINT	PK identity
notification_id	BIGINT	FK notifications.id
user_id	BIGINT	NN FK users.id
channel	VARCHAR(60)	NN; stores lookup value_code as raw string
provider_message_id	VARCHAR(150)	Nullable
delivery_status	VARCHAR(60)	NN; stores lookup value_code as raw string
sent_at	TIMESTAMPTZ	Nullable
delivered_at	TIMESTAMPTZ	Nullable
failure_reason	TEXT	Nullable

Constraints: Avoid logging secrets/provider credentials.

Indexes: (delivery_status_id,sent_at).

Lookup values: NOTIFICATION_CHANNEL; DELIVERY_STATUS

audit_logs

Purpose: Security and business audit history.

Column	Type	Constraints / notes
id	BIGINT	PK identity
actor_user_id	BIGINT	FK users.id
action	VARCHAR(100)	NN
entity_type	VARCHAR(60)	NN
entity_id	VARCHAR(100)	NN
old_values	JSONB	Nullable
new_values	JSONB	Nullable
ip_address	INET	Nullable

Constraints: Never log OTPs, access tokens, bank details, or sensitive document contents. Consider partitioning/retention.

Indexes: (entity_type,entity_id,creation_date DESC); (actor_user_id,creation_date DESC).

13. Table mappings and cardinalities

Parent / child	Mapping	Cardinality / rule
users → profile tables	users.id → *_profiles.user_id	1 to zero/one per profile type; names live in profiles
users ↔ roles	users.id → user_roles.user_id; roles.id → user_roles.role_id	Many-to-many
roles ↔ functions	roles.id → role_functions.role_id; functions.id → role_functions.function_id	Many-to-many, permissions on join table
users → addresses	users.id → addresses.user_id	One-to-many; intended for customer locations
partner → skills	partner_profiles.id → partner_skills.partner_id	One partner to many service skills
category → services	service_categories.id → services.category_id	One-to-many
service → catalogue children	services.id → service_variants.service_id / service_media.service_id / service_requirements.service_id / service_availability.service_id	One service to many variants, media records, requirements, availability rules and pricing rules
customer → bookings	customer_profiles.id → bookings.customer_id	One customer to many bookings
booking → assignments	bookings.id → booking_assignments.booking_id	Many assignment attempts; one accepted at a time
booking → history/timers	bookings.id → booking_status_history/booking_time_logs	One-to-many
booking → payments	bookings.id → payments.booking_id; payment_orders.id → payments.payment_order_id	Payment attempts; define split-payment behavior if needed
payment → refunds	payments.id → payment_refunds.payment_id	One payment to many refund requests
partner → payouts → transfers	partner_profiles.id → partner_payouts.partner_id; partner_payouts.id → partner_transfers.payout_id	Partner earnings and transfer attempts
booking → cancellation/review	bookings.id → booking_cancellations.booking_id /	At most one each in this design

	booking_reviews.booking_id	
ticket → messages/attachments	support_tickets.id → messages/attachments.ticket_id	One ticket to many
notification → delivery logs	notifications.id → notification_delivery_logs.notification_id	One notification to many delivery attempts
Lookup dropdowns	lookup_types.id → lookup_values.lookup_type_id; UI uses value_code/display_name; business columns store value_code as VARCHAR	No FK from business tables to lookup_values

14. Spring Boot service modules

Module	Responsibilities	Suggested service classes
Identity & access	Login, OTP, account, roles, function permissions	AuthService, OtpService, UserService, RoleService, PermissionService
Profiles & onboarding	Profiles, KYC, bank verification, training	ProfileService, PartnerOnboardingService, DocumentVerificationService
Catalogue & pricing	Categories, services, variants, , pricing and policies	CatalogService, PricingService, PolicyService
Availability & assignment	Schedule, leave, slot holds, partner eligibility, manual assignment	AvailabilityService, SlotReservationService, AssignmentService
Booking & timers	Cart checkout, booking lifecycle, service timer	CartService, BookingService, BookingTimerService
Payments & ledger	Orders, callbacks, refunds, commissions, payouts, cash	PaymentService, PaymentWebhookService, RefundService, CommissionService, PayoutService
Support	Tickets, messages and attachments	SupportService, AttachmentService
Notifications	In-app, push, SMS/email, retries	NotificationService, PushNotificationService, SmsService, EmailService
Audit & reporting	Admin audit, reports, operational history	AuditService, ReportingService
Lookups	Read/manage lookup types and values	LookupService

Recommended Java package structure

- com.yourcompany.app.config — CORS, OpenAPI, Jackson, scheduling and application config.
- com.yourcompany.app.security — security filter chain, token/session handling, authorization.
- com.yourcompany.app.common — exception handling, response envelope, pagination, audit base entity.
- com.yourcompany.app.lookup — lookup entities, repositories, services, controllers, seed migrations.
- com.yourcompany.app.identity — users, roles, user_roles, functions, role_functions, OTP.
- com.yourcompany.app.profile — customer/admin/partner profiles and addresses.
- com.yourcompany.app.catalog — categories, services, variants, service media, requirements, availability, FAQs and policies.
- com.yourcompany.app.partner — onboarding, documents, bank accounts, skills and availability.
- com.yourcompany.app.booking — cart, bookings, assignments, status history, timers and location.
- com.yourcompany.app.payment — orders, payments, webhooks, refunds, commission, payouts, cash.
- com.yourcompany.app.support; com.yourcompany.app.notification; com.yourcompany.app.audit.

15. Tools, dependencies and external services

Development prerequisites

Tool	Purpose / recommendation
JDK	Use a supported LTS version compatible with chosen Spring Boot; Java 21 is a reasonable baseline.
Maven or Gradle	Choose one build tool; Maven is straightforward for a new project.
PostgreSQL	Managed or self-hosted supported version; btree_gist extension if using exclusion constraints.
Flyway or Liquibase	Versioned database migrations and lookup seed values; choose one.
Node.js LTS + npm/pnpm	React and React Native tooling; follow React Native compatibility

	requirements.
Android Studio + Android SDK	Needed for local Android native builds/emulators; not required merely to write React components.
Git and API client	Git for source control; Postman/Bruno/Insomnia for API tests.
Docker (optional)	Local PostgreSQL/Redis and repeatable development environments.

Spring Boot dependencies

Dependency	Use
Spring Web	REST APIs
Spring Security	Authentication, authorization, method security
Spring Data JPA + PostgreSQL JDBC	ORM/repositories and database driver
Bean Validation	Request DTO validation
Flyway or Liquibase	Migrations and lookup seeding
Spring Boot Actuator	Health and metrics endpoints
Springdoc OpenAPI	API docs/testing; restrict in production
Spring Boot Mail (optional)	Email integration
Resilience4j or Spring Retry (optional)	Timeouts/retries for external APIs
Redis (optional)	Cache, rate limiting, short-lived coordination; not source of truth
Spring Scheduler or Quartz	Scheduled maintenance jobs; make jobs safe across multiple instances
Testcontainers (test scope)	Integration tests against real PostgreSQL

React Admin Dashboard

- React + TypeScript + Vite for the admin single-page app.
- React Router for navigation; TanStack Query for server-state fetching/cache.
- React Hook Form + Zod for form validation.
- A maintained component library such as MUI or Ant Design; server-side grid/table for admin lists.
- Use function permissions to show/hide screens, but always enforce authorization in Spring Boot.

React Native Customer and Partner apps

- React Native + TypeScript; choose Expo or Community CLI according to native SDK needs.
- React Navigation; TanStack Query; React Hook Form + Zod.
- Secure storage for tokens; do not put access tokens in plain AsyncStorage.
- Firebase Cloud Messaging (FCM) or compatible push setup.
- Maps/geocoding SDK for address selection and partner tracking; request location permissions only when needed.
- Camera/document upload support for partner onboarding; restrict file size/type and upload to protected storage.

External services to select

Capability	Options / decision criteria	Integration pattern
Payments & partner transfers	Compare Cashfree Easy Split and Razorpay Route; verify current marketplace onboarding, KYC, fees, settlement, refunds and partner transfer requirements before choosing.	Backend creates orders; signed webhooks update payments; backend calculates commissions and initiates transfers.
OTP/SMS	Indian transactional SMS provider; verify DLT registration/template compliance, delivery reports and rate limits.	Backend sends OTP and stores only hash, purpose, expiry and attempts.
Push notifications	Firebase Cloud Messaging is common for Android.	Store device tokens securely; log provider delivery status.
Email	Transactional email provider or managed SMTP; configure SPF/DKIM/DMARC.	Receipts, support updates and account messages.
Maps/geocoding	Google Maps Platform or another provider that supports target area and required SDKs.	Address lookup and authorized live tracking; review usage pricing/quotas.
File/object storage	S3, Azure Blob, Google Cloud Storage or compatible provider.	Private KYC/support files; use short-lived signed URLs.
Monitoring	OpenTelemetry-compatible logs/metrics/tracing; frontend error tracker such as Sentry.	Redact tokens, OTPs, bank details and personal documents.
Hosting/secrets	Cloud provider, managed PostgreSQL, secret manager/KMS.	TLS, private networking, backups, least-privilege credentials.

16. Suggested REST API groups

Area	Example endpoints
Authentication	/api/v1/auth/otp/request; /api/v1/auth/otp/verify; /api/v1/auth/logout
Current user/profile	/api/v1/me; /api/v1/customer/profile; /api/v1/partner/profile
Roles/permissions	/api/v1/admin/roles; /api/v1/admin/functions; /api/v1/admin/role-functions
Lookups	/api/v1/lookups/{typeCode}; admin-only lookup maintenance
Catalogue	/api/v1/categories; /api/v1/services; /api/v1/services/{id}/variants
Addresses	/api/v1/addresses
Partner onboarding	/api/v1/partner/documents; /api/v1/admin/partners/{id}/verify
Availability/assignment	/api/v1/partner/availability; /api/v1/admin/bookings/{id}/assign
Bookings	/api/v1/bookings; /api/v1/bookings/{id}; /api/v1/bookings/{id}/cancel
Service OTP	/api/v1/bookings/{id}/service-start/verify; /api/v1/bookings/{id}/service-completion/verify
Payments	/api/v1/payments/orders; /api/v1/payments/webhooks/{provider}; /api/v1/payments/{id}/refund
Support/notifications	/api/v1/support/tickets; /api/v1/notifications

These are proposed endpoint names, not a finalized API contract. Add DTO validation, pagination, authorization, rate limiting, consistent error responses and idempotency for payment/booking actions.

17. Security and implementation checklist

- Never store plaintext OTPs, passwords, access tokens, bank account details or provider secrets in logs.
- Use FK ON DELETE RESTRICT for transaction records and deactivate users/services/lookups rather than deleting referenced data.
- Use transactions for booking creation, slot reservation, payment updates, refunds, commissions and payout ledger changes.
- Verify provider webhook signatures and deduplicate events by provider + provider_event_id.
- OTP verification must check purpose, expiry, attempts, booking, assigned partner and current booking state; consume OTP atomically.
- Enforce role/function authorization on backend endpoints and service methods, not only by hiding frontend screens.
- Use numeric decimal types for money; snapshot service name, address, price and policy at booking time.
- Set retention rules for OTPs, location updates, webhook payloads and audit logs; redact sensitive fields.
- Back up PostgreSQL, test restores, enable TLS, manage secrets outside source control and use least-privilege credentials.

18. Recommended build order and table count

1. Create lookup_types and lookup_values for UI dropdown options, then seed codes via Flyway/Liquibase. Business tables store selected value_code strings without lookup foreign keys.
2. Create users, roles, user_roles, functions, role_functions and auth_otp_requests; add booking_id FK after bookings exists.
3. Create profile tables and partner onboarding tables.
4. Create service catalogue, pricing, FAQs and policy tables.
5. Create partner skills, availability, leave and slot reservations.
6. Create cart, bookings, assignments, timers and location updates.
7. Create payment orders, payments, webhooks, refunds, commissions, payouts, transfers and cash collection.
8. Create cancellation, reviews, support, notifications and audit tables.
9. Build and test each role in the React Admin and React Native apps; test authorization failures and payment webhooks.

Target count: 56 business tables plus lookup_types and lookup_values = 58 tables. Lookup dropdown values are stored as raw value_code strings in business tables, without foreign keys to lookup_values. Lookup dropdown values are stored as raw value_code strings in business tables, without foreign keys to lookup_values.

Lookup dropdown mapping: lookup_types and lookup_values define available UI options. Business tables store the selected value_code as a raw string and do not reference lookup_values with a foreign key. Validate each submitted code against its expected lookup type in the service layer.